

Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Status of the blind analysis and a request to open the control regions

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$\nu_\mu + \bar{\nu}_\mu$ CC single charged pion on argon • FHC, RHC and combined • blind, fake-data results

What is measured

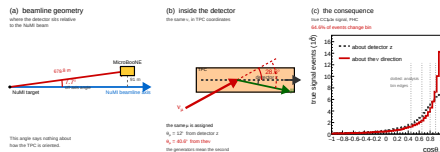
Signal (truth level, charge-blind): $\nu_\mu/\bar{\nu}_\mu$ CC in the FV with exactly one μ , exactly one $\pi^\pm$, no π^0 , no kaons or other mesons, any number of nucleons; coherent production included.

Measured phase space: $p_\mu > 0.15$, $p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad

Deliverables per configuration (FHC, RHC, combined)

- ▶ total flux-averaged cross section (cut-and-count)
- ▶ $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$ and $d\sigma/d\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, two-bin $d\sigma/dp_\pi$
- ▶ proton-tagged subsample: integrated cross section (headline); TKI and W_{had} as secondary results

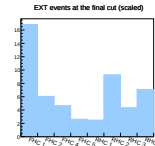
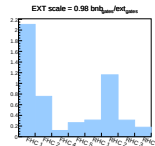
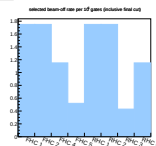
Blind analysis: every “data” point in this talk is Poisson-thrown central-value MC. No beam-on signal region has been opened.



MicroBooNE sits 7.7° off the NuMI axis; the beam arrives 28° from detector z, so all angles and transverse quantities are computed in the beam frame.

Exposure, normalisation and beam-off

Configuration	Runs	Data POT (10^{20})
FHC	1, 2, 4, 5	8.857
RHC	1, 2, 3, 4a-c	11.082
Combined	all 14 overlay files	19.939



- ▶ every run scaled by its own data/MC POT;
combined = per-mode data exposures
- ▶ per-configuration integrated flux (new-G4,
 $E_\nu > 60$ MeV): FHC 6.81, RHC 6.45, comb.
 $6.61 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$
- ▶ $N_{\text{Ar}} = 8.71 \times 10^{29}$ (dead z slab removed)
- ▶ CV weight = tune $\times$ PPFX $\times$ normalisation,
one safe-weight rule everywhere

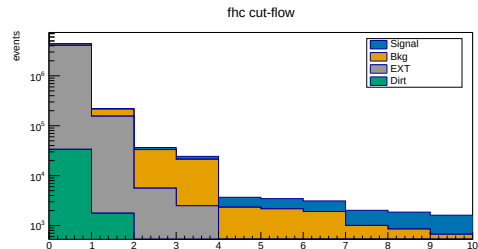
Beam-off: cosmic-only, matched by run period (horn label irrelevant), each run scaled by its beam-on/beam-off *gate* ratio. The pre-CRT Run-1 rate is 3–4 $\times$ the CRT-era one. EXT is 1.9% of the final sample (30 FHC events).

Selection and yields

Cuts: software trigger, vertex in FV, topological score > 0.67 , muon candidate (highest muon-BDT score among qualifying tracks), exactly one contained pion candidate (LLR, MIP-BDT, pion-BDT, Bragg- π), three-plane hits, shower veto, $\theta_{\mu\pi} < 2.6$ rad, non-proton multiplicity < 4 and < 5 tracks.

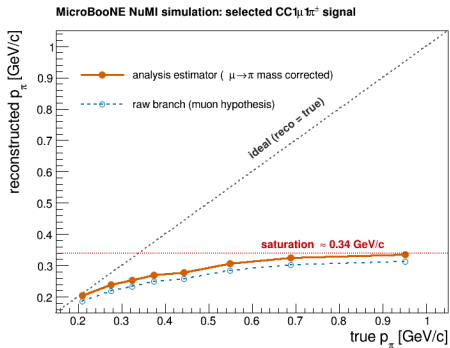
Final cut	Signal	ν bkg	EXT	Pred.	Pur.
FHC	943	637	30	1 612	58.5%
RHC	1 015	720	23	1 759	57.7%
Comb.	1 959	1 357	53	3 372	58.1%

Efficiency 17.4–17.6% in the measured phase space; largest background: CC0 π with a proton taken as the pion.

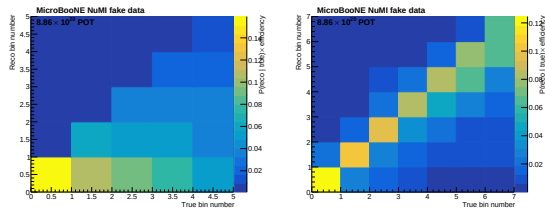


FHC cut-flow, per-run POT-scaled, full CV weight.

The pion momentum sets the scope



Reconstructed p_π (range under the muon hypothesis, mass-corrected) saturates near 0.34 GeV/c: pions interact before stopping.



Smearacceptance $S = \varepsilon M$: five-bin p_π (diagnostic, left) vs. p_μ (right).

- ▶ every estimator tried (golden-pion tag, MCS, calorimetry, regression, daughter recovery): a physics limit, not an algorithmic one
- ▶ p_π released as **two bins**: $[0.175, 0.205]$ and > 0.205 GeV/c, migration diagonals 91% / 76%
- ▶ proton-tagged $W_{\pi p}$ shape **withdrawn**; the sample supports an integrated cross section

Extraction: Wiener-SVD and what its integral means

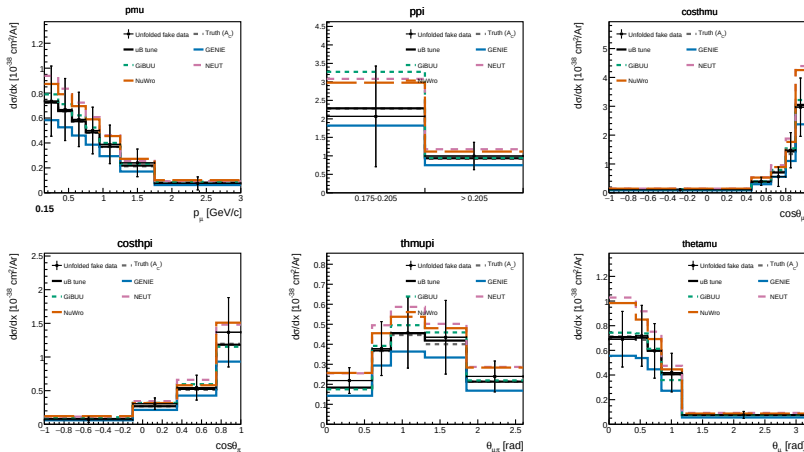
- ▶ $n^{\text{reco}} = S N^{\text{true}} + b$ with $S = \varepsilon M$ inverted once (no extra $1/\varepsilon$)
- ▶ Wiener-SVD (XSECANALYZER), second-derivative regularisation aware of the non-uniform bin widths
- ▶ the result is $A_C \times_{\text{true}}$; every prediction is smeared by the released A_C before comparison
- ▶ A_C is *not* norm preserving: the integral of a differential result is observable-dependent ($0.57\text{--}0.86 \times 10^{-38} \text{ cm}^2/\text{Ar}$); D'Agostini gives a flat, $\sim 30\%$ higher integral
- ▶ the physical total is the cut-and-count value

Total flux-averaged cross section ($10^{-38} \text{ cm}^2/\text{Ar}$)

	inclusive	proton-tagged
FHC	1.03 ± 0.43	0.63 ± 0.31
RHC	0.93 ± 0.42	0.55 ± 0.21
Comb.	0.98 ± 0.40	0.59 ± 0.25

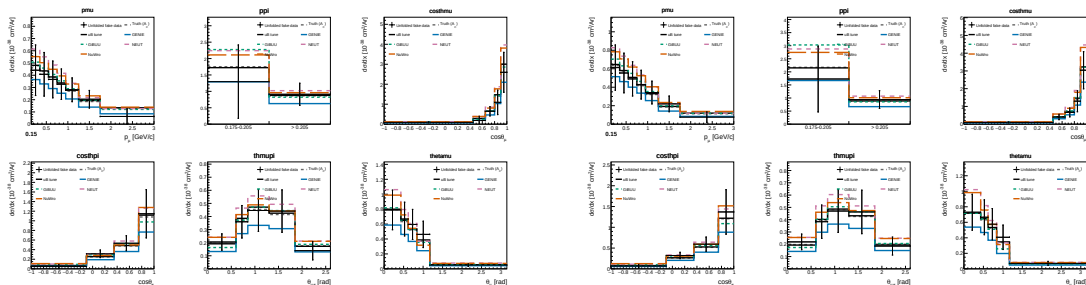
On fake data this *is* the injected tune (closure = 1 by construction); the uncertainty is what real data will carry. RHC/FHC = 0.90. GENIE v3 0.5–0.6 σ below, GiBUU/NEUT/NuWro within 0.7 σ .

Differential results, FHC



Unfolded fake data, A_C -smeared truth, tune and the four generators, all in the A_C -smeared space. Generators differ by normalisation ($\times 1.6$, GENIE to NEUT); shapes agree within the uncertainties. Full covariance: flux, cross section, re-interaction, detector, POT, targets, statistics.

RHC and combined

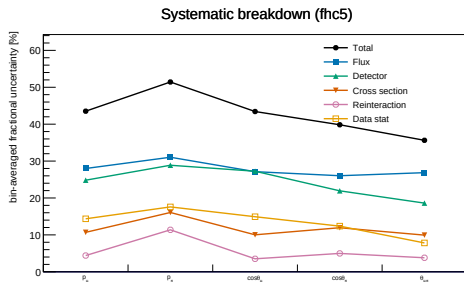


RHC

Combined FHC+RHC

Same 600 PPFX universes index-matched across both horn modes (rigorous flux correlation); native Run-4 RHC detector variations; each mode normalised to its own exposure and flux.

Uncertainties



Bin-averaged fractional uncertainty, inclusive (%):

Source	range	
Flux (PPFX)	24–34	dominant
Detector	17–43	~ flux for several obs.
Cross section	9–18	
Re-interaction	3–12	
Data statistics	7–20	
Total	34–62	

- ▶ the $\sim 17\%$ reco-level flux is amplified by the unfolding operator; the lever is response conditioning, not the flux model
- ▶ all 51 extractions carry a full covariance including detector variations; A_C and covariances released as text

Validation, and what the review found

Validation

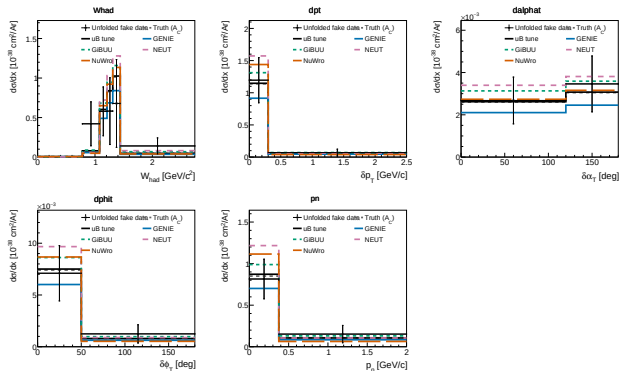
- ▶ fake-data closure: unfolded/realised truth = 1.016 on average over the 18 inclusive extractions; all $p > 0.8$
- ▶ 32-throw ensemble: pull widths 1.04 (p_μ), 0.96 ($\cos\theta_\mu$), 68/95% coverage recovered; 2.5% negative bias stated, not corrected
- ▶ clean-cache rebuild of all 51 extractions is bit-identical
- ▶ five automated guards: normalisation manifest, systematics/file coupling, staleness, table consistency, blinding guard

Corrected since August (all in the change log)

- ▶ beam-off scaled by an *event* count instead of gates: EXT was $9.7\times$ too large ($128 \rightarrow 30$ FHC events); now per run period
- ▶ fake-data truth carried the CV weight twice (1.6% on closure)
- ▶ PPFX CV weight missing from count-level tables (3–6%)
- ▶ one safe-weight rule everywhere; fake data re-thrown with it (1%)
- ▶ A_C applied twice to models; TKI computed about detector z (both withdrawn and redone)

None of these touched the real-data path; several would have biased a real-data result.

Proton-tagged subsample

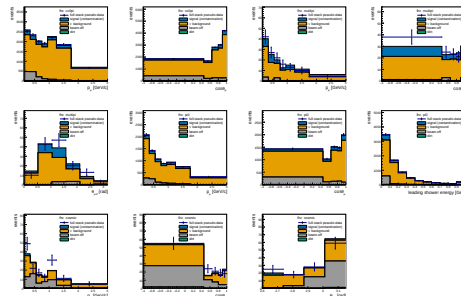


- ▶ inclusive selection + a contained proton $> 0.3 \text{ GeV}/c$; efficiency 11.4%, purity $\sim 50\%$; strong cosmic rejector
- ▶ TKI ($\delta p_T, \delta \alpha_T, \delta \phi_T, p_n$) in two bins, beam frame; W_{had} in six
- ▶ single-nucleon STV prescription applied to the πp system: p_n is a proxy (stated limitation)
- ▶ generators: normalisation spread only; no FSI discrimination at this exposure
- ▶ headline: integrated total 0.63 ± 0.31 (FHC)

Control regions

Region	Inversion	Role; $T = N^{\text{SR}}/N^{\text{CR}}$ of the target class
CC0 π	$n_\pi = 0$	tests 0 π feed-in; $T = 0.03$, $\delta T/T = 31\%$ (detector-limited)
multi- π π^0	$n_\pi \geq 2$ shower veto fails	diagnostic only; $T = 2.5$ π^0 leakage; $T = 0.02$, $\delta T/T = 15\%$: strongest handle
cosmic	$\theta_{\mu\pi} > 2.6$ rad	beam-off norm. and shape; 41% EXT purity, no fitted correction

- ▶ disjoint from the signal region event by event; multi- π and π^0 overlap ($\sim 40\%$ of multi- π)
- ▶ pseudo-data/prediction 0.99–1.10 everywhere; full-stack test (MC+EXT+dirt) at unity
- ▶ per-region systematics incl. detector variations; transfer factors as correlated ratios (flux cancels to 2–3%)



Frozen distributions and binning, FHC, full-stack pseudo-data in place of beam-on points.

Request to the collaboration

Open the four inclusive control regions; the signal region and the proton-tagged regions stay blind

Frozen before opening: selections at branch level, run-dependent normalisation, beam-off scaling, displayed observables and binning, covariance, goodness-of-fit and the permitted responses.

Procedure (to be ratified)

- ▶ FHC and RHC separately, combined as cross-check; data vs. MC+EXT+dirt
- ▶ pass: normalisation within 2σ (pre-fit predictive uncertainty) and shape $p > 0.05$
- ▶ allowed: no change; added uncertainty; one-parameter normalisation of the target class (purity $> 50\%$, LR test, transfer uncertainty added); else model revision and renewed approval
- ▶ forbidden: any signal cut or binning change; generator choice after inspection; bin-by-bin reweighting

Prerequisites

- ✓ tables from one release, checked automatically
- ✓ definitions, overlap = 0, plots and binning frozen
- ✓ EXT gates released per run; pairings asserted
- ✓ real beam-on files refused without XSEC.UNBLIND=1
- ✓ full-stack closure; region systematics incl. detector; transfer uncertainties
- ☐ ratification of the procedure

Before the signal region: PPFX-corrected flux constants confirmed; discrepancy treatment; clean-cache reproduction.

Summary

- ▶ A blind, fully validated extraction of $\nu_\mu + \bar{\nu}_\mu$ CC1 $\pi^\pm$ cross sections on argon in the NuMI beam: total and six differential distributions in FHC, RHC and combined, with full covariances and released A_C matrices.
- ▶ Two physics limits set the scope: the pion range saturates (two-bin p_π ; no $W_{\pi p}$ shape), and A_C is not norm preserving (the total is cut-and-count).
- ▶ Fake-data totals $1.03/0.93/0.98 \times 10^{-38} \text{ cm}^2/\text{Ar}$ with $\sim 41\%$ uncertainty, flux- and detector-limited.
- ▶ A sequence of normalisation and weighting defects was found by review and fixed; automated guards now stop each of them from recurring.
- ▶ We ask to open the four inclusive control regions under the frozen procedure.

Documents: analysis note (Draft 1.3), technical supplement, change log; data release in `report/data_release/`.